

Question Number	Answer	Additional guidance	Mark
7(a)(i)	An answer that makes reference to three of the following points: • as age increases the number of cases (of cancer in both sexes) increase (1) • (the number of cases in males continues to increase up to age 80) whereas the number of cases in females decreases {between 70 to 80 / after 79} (1) • males are more likely to get this cancer / converse for females (1) • conclusion as to why there is a difference between males and females (1)	accept more cases (of cancer) in older people accept there is a higher number of cases (of cancer) in all ages (after 30/ 39 / 40} accept this cancer is only found in people aged {over 39 / 40+} accept (males have greatest number after 80) whereas females have greatest between 70-79 (years) accept largest increase in cases is for males ages 60-69 accept more males have this cancer (than females) / converse e.g. idea that cancer allele could be on X chromosomes / inheritance (of this cancer) is sex-linked / other valid conclusion as to why males are more at risk e.g. hormone / lifestyle	(3)

Question Number	Answer	Additional guidance	Mark
7(a)(ii)	A calculation in which: • calculation of 3% of 2 million (1) • calculation of number of males with cancer (1)	<u>Example of calculation</u> $(0.03 \times 2\,000\,000) = 60\,000$ $(60\,000 \div 100\,000) \times 660 = 396$ no ecf correct answer scores full marks	(2)

Question Number	Answer	Additional guidance	Mark
7(b)(i)	• to allow the {chromosomes / chromatids} to be visible / in order to distinguish the phase of mitosis (1)	accept see which cells are in mitosis ignore to see which cells are dividing	(1)

Question Number	Answer	Additional guidance	Mark
7(b)(ii)	An explanation that makes reference to the following points: • count the number of cells in mitosis (1) • divide by the total number of cells (in tissue sample and $\times 100$) (1)	accept count number of cells with visible {chromosomes / chromatids} ignore count actively dividing cells	(2)

Question Number	Answer	Mark
7(c)(i)	The only correct answer is C prophase <i>A is not correct because chromosomes do not condense in anaphase</i> <i>B is not correct because chromosomes do not condense in metaphase</i> <i>D is not correct because chromosomes do not condense in telophase</i>	(1)

Question Number	Answer	Additional guidance	Mark
7(c)(ii)	An answer that makes reference to the following points: - preventing the condensing of chromosomes would {prevent mitosis occurring / reduce number of cells in mitosis / prevent prophase / prevent metaphase} (1) (reduced number of cells in mitosis) would {reduce the / lead to a low} mitotic {rate / index} (1) - therefore patients with cancer have a higher percentage survival (1)	accept preventing the condensing of chromosomes will {reduce mitosis / mean cells remain in interphase} accept takes longer to form cancer cells / fewer cancer cells produced accept patients live longer / smaller decrease in percentage survival	(3)